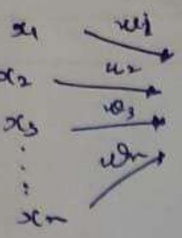
ANN and DL.



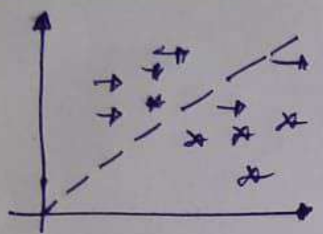
$\sum_1 x_i w_i + bias$ (z) $\longrightarrow f(z)$ $\longrightarrow$ Output

[Activation function]

Predicted = y
Actual = $\hat{y}$

Loss $\hat{y} - y$:→ To reduce the loss use Back propagation by changing the weights.

Linear Model of Neuron:-



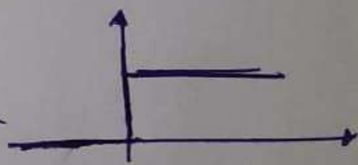
$w^T x + b > 0$

$w^T x + b < 0$

$w^T x + b = 0$

Activation function:- defines the output of a neuron.

$$\phi(v) = \begin{cases} 1, & v \geq 0 \\ 0, & v < 0 \end{cases}$$ } Threshold function



Piece Wise function:-

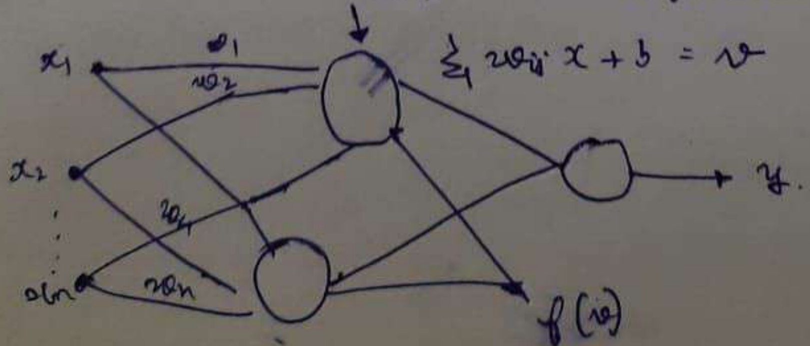
$\phi(v)$:

$$\text{Tanh hyperbolic} = \frac{e^z - e^{-z}}{e^z + e^{-z}} \in [-1, +1].$$

Which to use ??

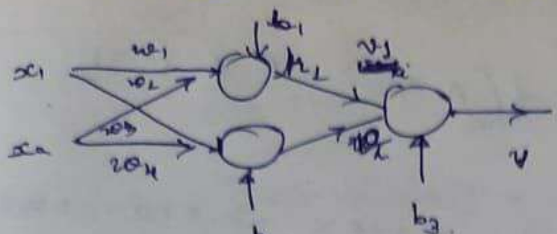
↳ Hidden layers ReLu, leaky ReLU, GELU.

↳ Output layers Binary [Sigmoid]
Multiclass [Softmax]
Regression → X needed.



Computation Node :→ activation function + summation.

Communication Link→ Indicate direction (Input → Output)



$v = h_1 v_1 + h_2 v_2$

Activation function → Non Linear.

Rule 1

Synaptic link :→ Linear summation function.

Eg:- $x_1 = 1$ $x_2 = 2$. $w_1 = 0.5$ $w_2 = 0.2$.

Synaptic → (Linear) $z = 0.5 \times 1 + 0.2 \times 2 = \underline{0.9}$.

$y = \text{Relu}(0, 0.9) = \underline{0.9}$.

Node signal = $\sum_i$ signals along

$y_{total} = y_1 + y_2$.

Rule 2

Eg:-

Input	weight	weighted-input
$x_1 = 1$	$w_1 = 0.4$	0.4
$x_2 = 2$	$w_2 = 0.6$	1.2
$x_3 = -1$	$w_3 = 0.5$	-0.5

~~weighted~~ use activation f^n → Relu.

Synaptic

} Relu = $\underline{1.1}$.

Rule 3 neuron A → output y_A and B and C with w_{BA}, w_{CA}.

Input ~~to~~ B : $x_B = w_{BA} y_A$

$x_C = w_{CA} y_A$

Eg:-

outgoing link	Weight	Signal out
To N_1	0.5	0.8×0.5
To N_2	0.2	0.8×0.2
To N_3	1.0	0.8×1.

} Assume input to this is 0.8 (output of previous)

Eg:- Simple recurrent Computation :-

$y(n) = 0.5\, x(n) + 0.4\, y(n-1)$

with $x(1) = 2$ $y(0) = 0$.

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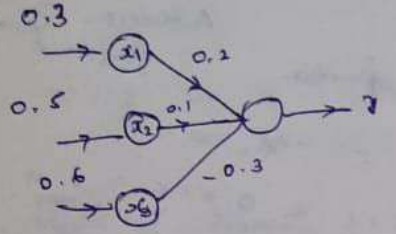
$y(1) = 0.5x(1) + 0.4y(0)$

$y(1) = 0.5 \times 2 + 0.4 \times 0 = \underline{1}.$

$y(2) = 0.5 \times 1$

$y_n = f\left(A\, x_i(n) + R\, y_n(n-1)\right).$

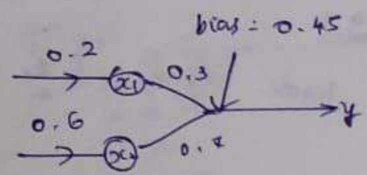
Eg:-



$z = 0.3 \times 0.2 + 0.5 \times 0.1 + 0.6 \times -0.3$

$= \underline{-0.07}.$

Eg:-



with bias

$z = 0.2 \times 0.3 + 0.6 \times 0.7 + 0.45$

$= \underline{0.93}.$

without bias :→

$z = 0.2 \times 0.3 + 0.6 \times 0.7$

$= \underline{0.47}.$

Eg:-

Input	x_i	w_i
I_1	0.5	0.4
x_2	0.2	0.3
x_3	0.8	0.6

bias = 0.1

$f(x) = \frac{1}{1+e^{-x}}$ (sigmoid)

} Single layer feed forward

Net input $= 0.5 \times 0.4 + 0.2 \times 0.3 + 0.8 \times 0.6 + 0.1 = \underline{0.84}.$

$f(x) = \frac{1}{1+e^{-0.84}}.$

Numerical Backpropagation:-

Eg: Inputs :- $x_1 = 0.05$ $x_2 = 0.10$

weights: $w_1 = 0.15$ $w_2 = 0.20$

bias :- $b = 0.35$

Target output: $t = 0.01$.

Learning rate $(\eta) = 0.5$.

Activation function :- sigmoid $= f(x) = \frac{1}{1+e^{-x}}$.

$v = w_1 x_1 + w_2 x_2 + b$

$v = 0.15 \times 0.05 + 0.20 \times 0.10 + 0.35 = \underline{0.3775}$

$f(v) = \frac{1}{1+e^{-0.3775}} = \underline{0.59321}$

Loss function [cost function] $= \frac{1}{2}$ (target − predicted)

Feedforward

$= \frac{1}{2}(0.01 - 0.593)^2 = \underline{0.170}.$

$\boxed{w_{new} = w_{old} + \text{learning rate} \times \text{derivative error} \times x_i}$

$\delta = (t-y) f'(v) \qquad f'(v) = y(1-y) = 0.593\,(1-0.593) = \underline{0.0241}$

$\delta = (t-y)y(1-y) = (0.01 - 0.593)\; 0.593\,(1-0.593) = \underline{-0.140}.$

~~$\Delta w = \eta \delta x_i$~~ $\quad w_{new} = w_{old} + \eta \delta x_i$

$w_{new} = 0.15 + 0.5 \times (-0.140) \times 0.05$

$w_1 = \underline{0.1465}.$

$w_2 = 0.10 + (0.10)(0.5)(-0.140) = \underline{0.093}.$

$w_1 = 0.1465, \quad w_2 = 0.193.$

2nd iteration $\quad x_1 = 0.05 \quad x_2 = 0.10$

$v = 0.05 \times 0.1465 + 0.1 \times 0.1930 + 0.35 = \underline{0.3766}$

$\Delta w = \eta \delta x_i$

$f(v) = \frac{1}{e^{-0.3766}} = \underline{0.5930}.$ $\quad$ error $= \frac{1}{2}(t - f(v))^2 = 0.\underline{1699}$

$= \frac{1}{2}(0.01 - 0.5930)^2$

Hebbian learning.

$\boxed{\Delta w = \eta\, y_i x_i}$

Adaptive filtering:

(i) Filtering process :→ $y(i) = \sum_1 w_i x_i^T \longrightarrow$ ①.

(ii). Adaptive process :→ $e(i) = d(i) - y(i) \longrightarrow$ ②.

$d(i)$ ↓ target; $y(i)$ ↓ obtained output

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$\boxed{\nabla E(w^*) = 0}$ → condition for optimality.

$$\nabla = \left[\frac{\partial}{\partial w_1}, \frac{\partial}{\partial w_2}, \frac{\partial}{\partial w_3}, \ldots \ldots \frac{\partial}{\partial w_n}\right]$$

$$\nabla E(w^*) = \frac{\partial E_1}{\partial w_1}, \frac{\partial E_2}{\partial w_2}, \ldots \ldots \frac{\partial E_n}{\partial w_n}$$

$$\boxed{w(n+1) = w(n) - \eta\, g(n)}$$

$$\Delta(w(n)) = w(n+1) - w(n) = -\eta\, g(n).$$
$$= w(n) - \eta\, g(n) - w(n)$$

$$\boxed{\Delta w(n) = -\eta\, g(n)}$$

LMS algorithm:- loss function: $E(w) = \frac{1}{2}\{e(w)\}^2$.

diff. w.r. to 'w', we get.

$$\frac{\partial E(w)}{\partial w} = \frac{\partial}{\partial w}\left(\frac{1}{2} e^2(w)\right)$$
$$= \frac{1}{2} \sum_1 \frac{\partial}{\partial w_i}\left(d(i) - y(i)\right)^2.$$
$$= \frac{1}{2} \times \sum_1 2\left(d(i) - y(i)\right) * \frac{\partial}{\partial w}\left(d(i) - y(i)\right)$$
$$= \sum_1 d(i) - y(i) * \frac{\partial}{\partial w}\left(d(i) - \sum_1 w_i x_i^T\right)$$

$$\nabla E(w) = \sum_1 \left(d(i) - y(i)\right) \times (-x_i)$$
$$\nabla E(w) = -x \times e(w)$$

$g(n) = \nabla E(w)$

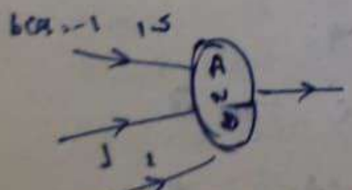
$$\Delta w(n) = -\eta\, g(n)$$
$$= -\eta \sum_1 \left(d(i) - y(i)\right) \times (-x_i)$$
$$= \eta \sum_1 \left(d(i) - y(i)\right) \times (-x_i)$$
$$\boxed{\Delta(w) = \eta\, e_i x_i}$$

Condition convergence

$$\boxed{0 < \eta < \frac{2}{\lambda_{max}}}$$

$$\boxed{\eta = \frac{c}{n}}$$ time varying

$$\eta = \frac{\eta_0}{\left(1 + \frac{n}{\tau}\right)}$$



$$-1 \times 1.5 + 0 \times 1 + 0 \times 1 = \underline{-1.5}.$$
$$\frac{1}{1 + e^{-1.5}} = 0.\underline{1624}.$$

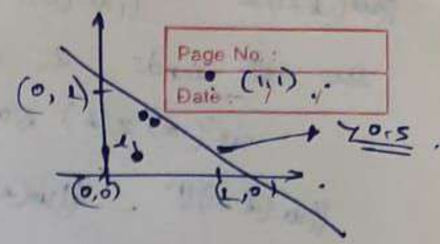
$$-1 \times 1.5 + 0 \times 1 + 1 \times 1 = -0.5$$
$$\frac{1}{1 + e^{-0.5}} = \underline{0.377}.$$

$-1 \times 1.5 \times 1 \times 1 + 0 \times 1 = \underline{-0.5}$

$\frac{1}{1+e^{0.5}} = \underline{0.377}$

$-1 \times 1.5 + 1 \times 1 + 1 \times 1 = \underline{0.5}$

$\frac{1}{1+e^{-0.5}} = \underline{0.622}$



Perception convergence theorem:-

$w^T x > 0 \rightarrow C_1$
$w^T x \leq 0 \rightarrow C_2$

For misclassified inputs.

$w(n+1) = w(n) - \eta \times \text{error}(n)$

If $w^T x > 0$ and $x \in C_2$ } actually C_1

If $w^T x \leq 0$ and $x \in C_1$ } actually C_2. $w(0) = 0$.

Given linear separable inputs, learning rate $\eta = 1$, weights $= 0$.

1. Assume that 1^{ST} 'n' examples $\in C_1$ are all classified, then using

$w(n+1) = w(n) + x(n)$.

$w(n+1) = x(1) + x(2) + \dots + x(n) \rightarrow$ ①.

2. Now we need to define +ve margin against the true separator.

$\alpha = \min_{x \in C_1} w_0^T x(n) > 0 \rightarrow$ ②.

multiply both sides in eq ① with w_0^T we get:-

$w_0^T \times w(n+1) = w_0^T x(1) + w_0^T x(2) + \dots + w_0^T x(n) \rightarrow$ ③

from ② and ③, we get,

$w_0^T w(n+1) > n\alpha \longrightarrow$ ④.

apply cauchy schwarz inequality $|a^T b| = \|a\| \|b\|$

$\|w_0\|^2 \|w(n+1)\|^2 \geq [w_0^T(n+1)]^2$

$w(n+1) = w(n) + x(n)$

Method 2:- taking euclidian distance of the sequence using the polarization identity.

Now expand

$\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2U^T V$.

applying $U = w(k)$ $V = x(k)$.

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$$\|w(k+1)\|^2 = \|w(k)\|^2 + \|x(k)\|^2 + 2 \times w(k)^T x(k).$$

Since all inputs in C_1 are misclassified.

$$\therefore \; w^T(k)\, x(k) \leq 0 \quad \forall \; k = 1, 2, 3, \dots n.$$

$$\|w(k+1)\|^2 - \|w(k)\|^2 - \|x(k)\|^2 = 2\, w^T(k) \times x(k) \leq 0.$$

$$\|w(k+1)\|^2 - \|w(k)\|^2 \leq \|x(k)\|^2.$$

Telescopic sum of β : $\sum_1^n a_k \leq n \max_k a_k$.

Now sum from $k = 1$ to n, $w(0) = 0$.

$$\|w(k+1)\|^2 \leq \sum_1^n \|x(k)\|^2 \leq n\beta \quad \text{where} \quad \beta = \max_{x \in C_1} \|x(k)\|^2$$

Now from equations,

$$\boxed{\frac{n^2 \alpha^2}{\|w_0\|^2} \leq \|w(n+1)\|^2 \leq n\beta}$$